

Sistemas Multiagentes (PPGCA-CT e PPGCC-PG)

Projeto Inicial (PI)

Prof^a Celia Ghedini Ralha

Universidade Tecnológica Federal do Paraná (UTFPR)
Curitiba/PR, Brasil

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Agenda



- O que é para fazer?
 - ▶ Elementos Básicos
 - ▶ O Que/Como/Quando entregar
- Modelagem Prometheus
- Example Navigation on 2D Grid (MASPY) (link Moodle)
<https://github.com/laca-is/MASPY/wiki/Navigation-on-2D-Grid>

Projeto Inicial: Elementos Básicos



- Especificar Task Environment (PEAS) - PP

- ▶ Performance measure
- ▶ Environment
 - Fully/Partially Observable
 - Deterministic/Nondeterministic-Stochastic
 - Episodic/Sequential
 - Static/Dynamic
 - Discrete/Continuous
 - Single/Multi-agent
- ▶ Actuators
- ▶ Sensors

- Definir Tipos de Agentes

- ▶ Simple Reflex Agent
- ▶ Reflex Agent with State
- ▶ Goal-based Agent
- ▶ Utility-based Agent
- ▶ Learning Agent

- Fazer a Modelagem (e.g., Prometheus) - MOD

- Implementar Walker com 2 targets usando exemplo Navigation on 2D Grid do MASPY (targets com valores diferentes, os agentes competem pelo > valor)

Projeto Inicial



O PI é em grupo e vale 20% da nota na disciplina

- O que entregar?
 - ▶ Especificação PEAS, Definir tipos de agentes e Modelagem Prometheus (arquivo PDF)
 - ▶ Implementação em MASPYPY (código fonte)
- Como e quando entregar?
 - ▶ Submeter via Moodle - Sistemas Multiagentes (PPGCA-CT e PPGCC-PG) <https://moodle.utfpr.edu.br/course/view.php?id=36307>.
 - ▶ A partir de 09/10 até 30/10/2026 (23:59)

Exemplo PEAS



Em uma rede de telecomunicações de alta velocidade, a coordenação em tempo real entre diferentes agentes de software garante que milhares de usuários naveguem sem interrupções. Imagine o cenário de um estádio de futebol lotado durante uma final de campeonato.

- Optical Line Terminal (OLT) - atua na borda da rede de fibra óptica do estádio e monitora o tráfego.
- Baseband Unit (BBU) - localizado no data center da operadora proximo ao estádio, gerencia o processamento dos sinais de rádio da rede móvel que atende a região.
- Resource Management Agent (RMA) - supervisiona a infraestrutura global da operadora, analisa os pedidos da BBU e os dados de fluxo da OLT.

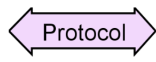
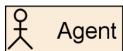
Prediction Agent (PA) PEAS

Component	Description
Performance Measure	Minimize demand prediction error.
Environment	OLT, BBU mobile scheduling data with characteristics: partially observable; non-deterministic; episodic; dynamic; continuous; and multi-agent.
Actuators	Sends predicted bandwidth demands to RMA.
Sensors	BBU scheduling interface (collects mobile packet information).

Prometheus: Modeling Notation Language¹

Para que serve o Prometheus?

- Assist developers to design, document, and build agent systems
- Used in designing BDI agent systems
- Generic to any MAS architecture and implementation environment
- Developed in collaboration with Agent Oriented Software (industry, academy)
- System overview diagram ties together the agents, events and shared data objects

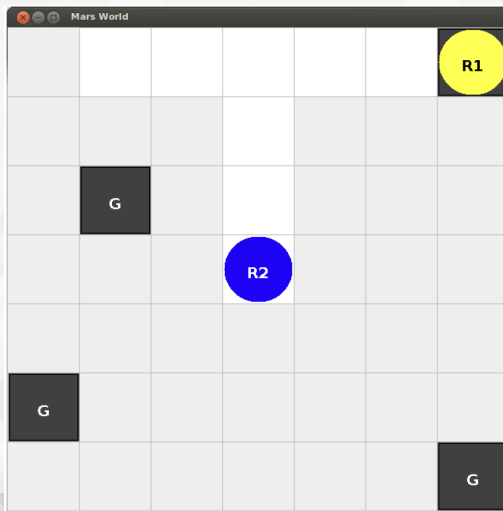


¹

- Lin Padgham & Michael Winikoff. Prometheus: a methodology for developing intelligent agents. In Proc. 1st International Joint Conference on Autonomous Agents and Multiagent systems (AAMAS'02). ACM, NY, USA, 37-38. <https://doi.org/10.1145/544741.544749>.

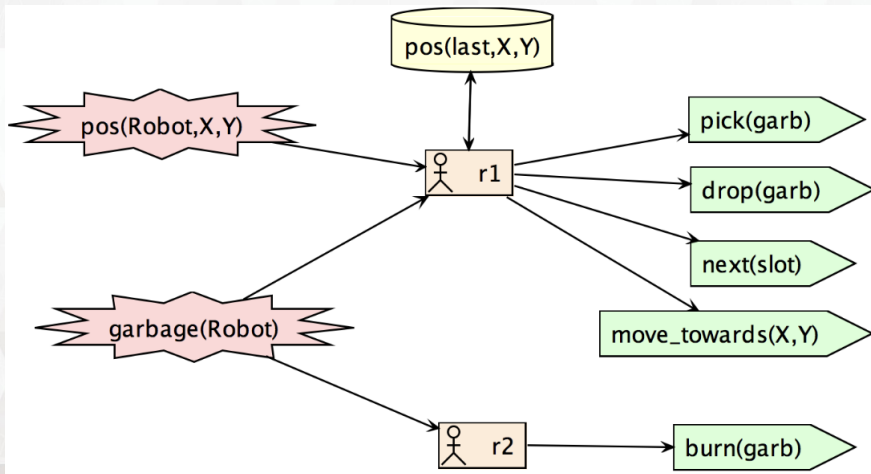
- John Thangarajah, Lin Padgham & Michael Winikoff. Prometheus design tool. In Proc. 4th International Joint Conference on Autonomous Agents and Multiagent systems (AAMAS'05). ACM, NY, USA, 127-128. <https://doi.org/10.1145/1082473.1082817>

Cleaning-robots Example



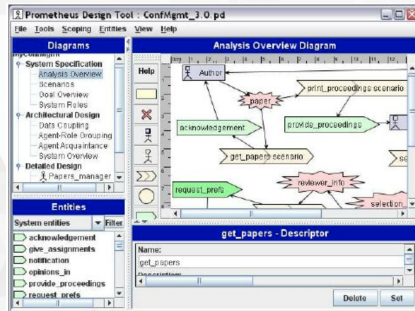
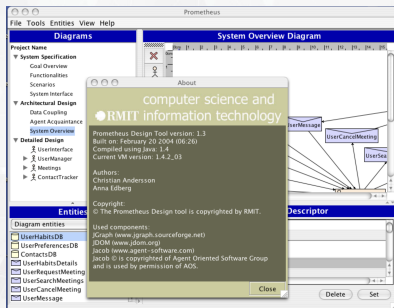
Disponível em: jason-bin-3.3.0/examples/cleaning-robots

Cleaning-robots Example: Prometheus



Prometheus Design Tool (PDT)

- PDT integrated into Eclipse platform²



- PDR foi distribuído aplicação Java independente (JAR executável/JRE) - procurar repositórios do GitHub

²

Sun, Hongyuan; Thangarajah, John; Padgham, Lin (2010). Eclipse-based prometheus design tool. Royal Melbourne Institute of Technology (RMIT) University. <https://hdl.handle.net/10779/rmit.27400377.v1> (descontinuada)

Ferramentas de Diagramação Online de Uso Geral



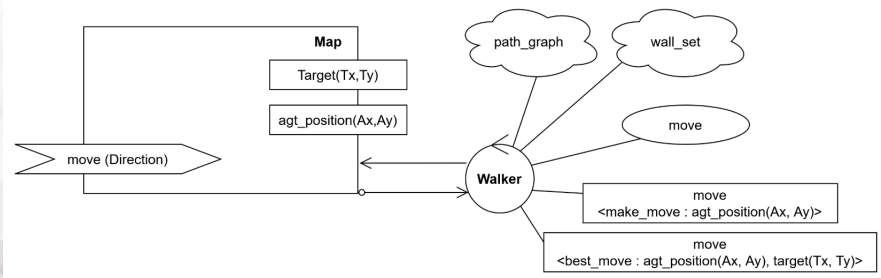
- Lucidchart: bibliotecas extensas de UML, formas customizadas e suporte a raias (swimlanes) ideais para mapear os protocolos de interação entre agentes.
- Miro: bom para a fase inicial de System Specification, precisa agrupar percepções, ações e metas de forma colaborativa com post-its e setas flexíveis.
- Draw.io/Diagrams.net: executa direto no navegador e permite criar formas personalizadas que você pode salvar como "componentes de agente"(e.g., a caixinha do Agente, do Plano, do Evento e do Protocolo)

Navigation 2D Grid Example

This example comprises n walker agents placed randomly in a 2D grid with configurable number of walls and dimensions.

The diagram presents the connections between agents & environment. Agents' actions through beliefs and goals.

- The Walker has: 2 Beliefs (path_graph and wall_set) + 1 goal (move) find the target + 2 plans (know target, doesn't know target)
- The Map has: 2 initial percepts (target, agent position)
- 1 Action: move in a specific direction



Navigation 2D Grid Example - Reproduction

Requirements

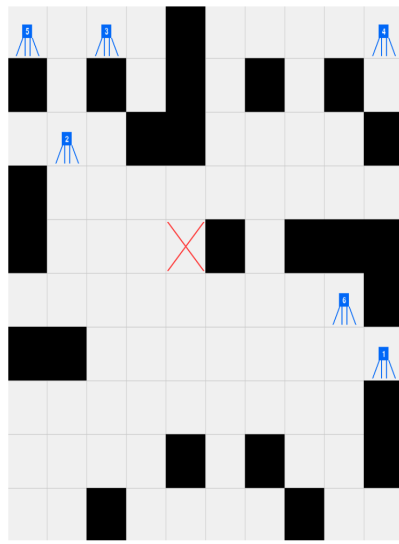
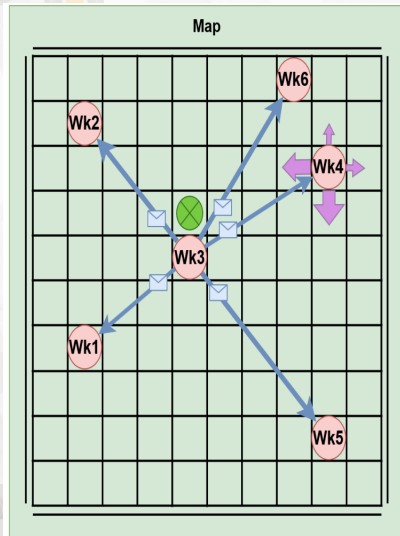
- Install

- ▶ MASPYP Framework
- ▶ MASPYP-GUI
- ▶ Pygame UI

- Download and run code

<https://github.com/laca-is/MASPYP/tree/main/examples/2D-Navigation-Learn>

Navigation 2D Grid Example - Visualização



The Walker



- Each walker trains to find the target, starting from their placed position.
- Walkers communicate exchanging information about the paths they took, the walls they found and the target location.
- Walkers can only perceive the position they are in.
- Walkers can see ahead using their internal method *filter_perceptions()* adding a focus to only perceive percepts with the agent name
- Walkers start with Path and Wall Beliefs empty with the Goal to move
- Its during the Walker initialization that the partial training happens

The Map



- Have 2 dimensional sizes + set number % of walls
- With a larger Wall percentage, an error can happen where the Map can be impossible to traverse completely
- When this occurs the Systems throws a error in *generate_map_with_walls()*
- Starts with a perception for the random target and for the walls
- Whenever a new Walker connects to the Map, it is assigned a random position
- This position is set to be at least 2 cells of distance of the target.

Thank you for coming!



Prof^a Celia Ghedini Ralha
celiaghedini@utfpr.edu.br